

Question				Marking details	Marks available																													
					AO1	AO2	AO3	Total	Maths	Prac																								
10/2	(a)	(i)		glowing splint (in the tube) – it relights	1			1		1																								
		(ii)		20		1		1		1																								
	(b)	(i)		<table border="1"><thead><tr><th rowspan="2">Electrolyte</th><th colspan="2">Ions present in the electrolyte</th><th colspan="2">Observations</th></tr><tr><th>Positive ion(s)</th><th>Negative ions(s)</th><th>At the negative (–) electrode</th><th>At the positive (+) electrode</th></tr></thead><tbody><tr><td>molten lead(II) bromide</td><td>Pb²⁺</td><td>Br[–]</td><td>grey metal A formed</td><td>orange gas formed</td></tr><tr><td>aqueous copper(II) chloride</td><td>Cu²⁺ and H⁺</td><td>Cl[–] and OH[–]</td><td>brown metal formed</td><td>green-yellow gas B formed</td></tr><tr><td>aqueous compound C</td><td>Zn²⁺ and H⁺</td><td>I[–] and OH[–]</td><td>grey metal formed</td><td>brown solution formed</td></tr></tbody></table> award (1) for each correct ion	Electrolyte	Ions present in the electrolyte		Observations		Positive ion(s)	Negative ions(s)	At the negative (–) electrode	At the positive (+) electrode	molten lead(II) bromide	Pb ²⁺	Br [–]	grey metal A formed	orange gas formed	aqueous copper(II) chloride	Cu ²⁺ and H ⁺	Cl [–] and OH [–]	brown metal formed	green-yellow gas B formed	aqueous compound C	Zn ²⁺ and H ⁺	I [–] and OH [–]	grey metal formed	brown solution formed	2			2		2
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		(ii)		metal A lead / Pb (1) do not accept Pb ²⁺ gas B chlorine / Cl ₂ (1) do not accept chloride / Cl [–] compound C zinc iodide / ZnI ₂ (1) accept ZnI			3	3		3																								
				Question 10/2 total	3	1	3	7	0	7																								